Functional Groups

Answer key with worked solutions

A. Understanding

1. Functional groups.

> (a) An **atom or group of atoms** that gives a family of organic compounds its typical reactions and properties. (b) Both contain the **same functional group (–OH)**; reactions happen at the functional group, and the different chain length has little effect.

B. Identifying and naming

2. Functional groups.

> (a) **alcohol** (b) **aldehyde** (c) **ketone** (d) **carboxylic acid** (e) **ester** (f) **amine** (g) **haloalkane** (chloro) (h) **amide**

3. Names.

> (a) **propan-1-ol** (b) **propanal** (c) **butanone** (butan-2-one) (d) **methanoic acid** (e) **propanoic acid** (f) **2-chloropropane**

4. Condensed formulas.

> (a) $CH_3CH(OH)CH_2CH_3$ (b) $CH_3CH_2CH_2CH_2CHO$ (c) $CH_3CH_2COOCH_3$ (d) $CH_3CH_2CH_2COOH$

C. Isomers and properties

5. Functional group isomers.

> (a) **propanal**, CH_3CH_2CHO (aldehyde), and **propanone**, CH_3COCH_3 (ketone). (b) **ethanol**, CH_3CH_2OH (alcohol), and **methoxymethane**, CH_3OCH_3 (ether).

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6. Boiling points.

> **Propane** has only weak dispersion forces. **Methoxymethane** is polar (C–O bonds), so it has dipole–dipole forces too, but no O–H bond, so no hydrogen bonding. **Ethanol** has an O–H group and forms **hydrogen bonds** between molecules, so much more energy is needed to separate them. **Ethanoic acid** has an O–H and a C=O group, forms **stronger and more hydrogen bonds** (molecules pair up), and is also heavier, so it boils highest.

D. Esters

7. Esters.

> (a) **methyl ethanoate**: $CH_3COOH + CH_3OH \rightleftharpoons CH_3COOCH_3 + H_2O$ (b) **ethyl propanoate**, $CH_3CH_2COOCH_2CH_3$ (c) **methanoic acid** ($HCOOH$) and **propan-1-ol** ($CH_3CH_2CH_2OH$)

8. Yield of methyl ethanoate.

> (a) $n(CH_3OH) = \dfrac{4.00\ \text{g}}{32.04\ \text{g/mol}} = 0.1248\ \text{mol}$; the ratio is $1:1$, so $n(\text{ester}) = 0.1248\ \text{mol}$
> $m = 0.1248\ \text{mol} \times 74.08\ \text{g/mol} = \mathbf{9.25\ g}$
> (b) Percent yield $= \dfrac{6.50\ \text{g}}{9.25\ \text{g}} \times 100\ \% = \mathbf{70.3\ \%}$. The reaction is **reversible** and reaches equilibrium, so not all the methanol is converted (some product is also lost when it is separated and purified).

E. Challenge

9. Identifying the liquid.

> (a) In 100 g: $n(\text{C}) = \dfrac{54.53\ \text{g}}{12.01\ \text{g/mol}} = 4.540\ \text{mol}$; $n(\text{H}) = \dfrac{9.15\ \text{g}}{1.008\ \text{g/mol}} = 9.077\ \text{mol}$; $n(\text{O}) = \dfrac{36.32\ \text{g}}{16.00\ \text{g/mol}} = 2.270\ \text{mol}$
> Divide by the smallest (2.270 mol): C 2.000, H 3.999, O 1, so the empirical formula is C_2H_4O (44.05 g/mol).
> $\dfrac{88.1\ \text{g/mol}}{44.05\ \text{g/mol}} = 2.00$, so the molecular formula is $C_4H_8O_2$.
> (b) $C_4H_8O_2$ with a sweet smell and no reaction with carbonate: an **ester**, for example **ethyl ethanoate**, $CH_3COOCH_2CH_3$ (methyl propanoate and propyl methanoate are also correct).

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